

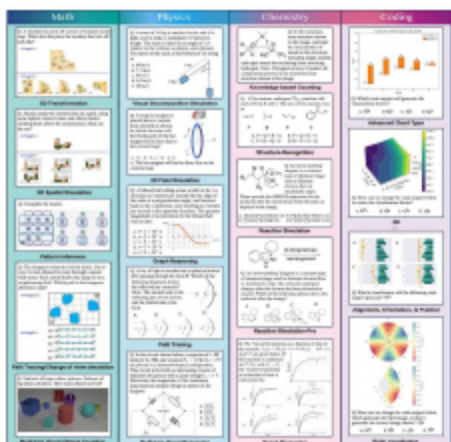
EMMA: An Enhanced MultiModal Reasoning Benchmark

Yunzhuo Hao et al. • UESTC, SYSU, UW, Microsoft, CUHK • 2025

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Why Multimodal Reasoning Matters

3 Landscape of Multimodal STEM Tasks



- Covers Math, Physics, Chemistry, and Coding
 - Mix of diagrams, plots, code, and symbolic content
 - Requires perception plus stepwise reasoning
- Questions vary by domain; formats mostly MCQ
 - Integrates text and visuals for grounded inference
 - Tasks span qualitative and quantitative analysis

4 Limitations of Benchmarks & Need for EMMA

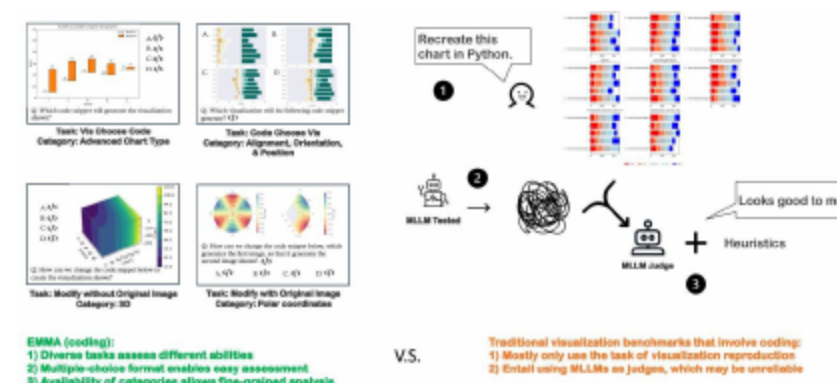
- Many tests judge visualization, not reasoning accuracy.
- Coverage is narrow; multi-hop and spatial skills underused.
- Annotations are inconsistent or noisy; little human vetting.
 - Overlooks deep visual reasoning; favors shallow perception.
 - EMMA introduces diverse, reasoning-centric STEM MCQs.
- Tasks demand integrated visual-language problem-solving.

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EMMA: What, How & What's New

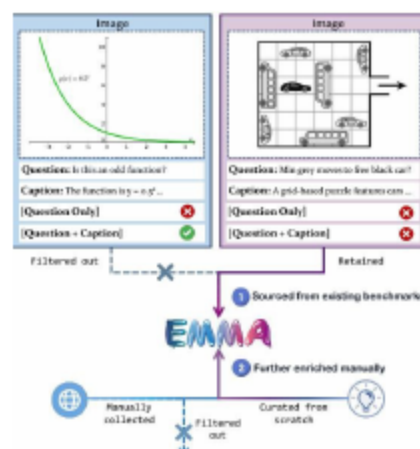
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EMMA vs. Visualization Benchmarks



- EMMA focuses on reasoning, not visualization.
 - 2,788 MCQ tasks across math, physics, chemistry, coding.
 - Rigorous filtering ensures true visual reasoning.
- Answers are human-checked for correctness.
 - Standardized evaluation enables fair comparisons.
 - Positions EMMA vs. visualization reproduction tasks.

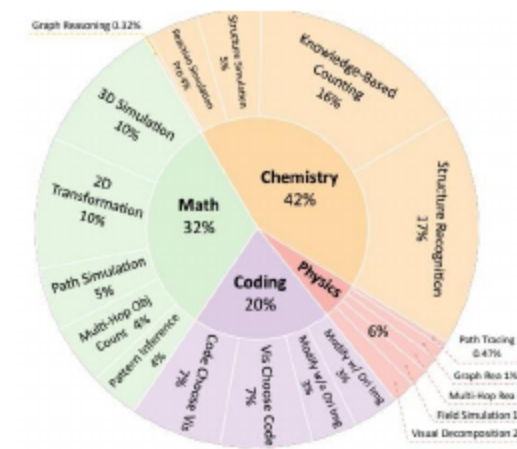
7 EMMA Data Curation & Quality Control



- Aggregate sources across Math, Physics, Chemistry
 - Caption-guided retention filters weak visuals
 - Experts manually craft and enrich challenging MCQs
- Ambiguous or low-signal items removed early
 - Consistency checks ensure single, reliable answers
 - Focus on visual reasoning, not rote recall

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Benchmark Composition & Task Taxonomy



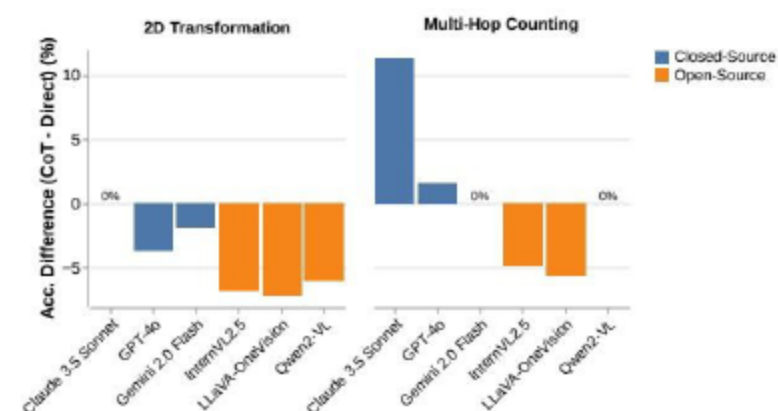
- 2,788 tasks spanning math, physics, chem, coding
 - Domain mix: Chem 42%, Math 32%, Coding 20%, Phys 6%
 - Tasks span counting, structure recognition, transformations
- Balanced difficulty levels across domains and skills
 - Rigorous filtering ensures truly visual reasoning
 - Improves on prior sets via stricter screens, new tasks

EXP

Experiments: Eval, Results, Scale

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Evaluation & Headline Results



• MCQ evaluation with fixed answer choices.

– Closed-source lead; big gaps on 2D transform and multi-hop.

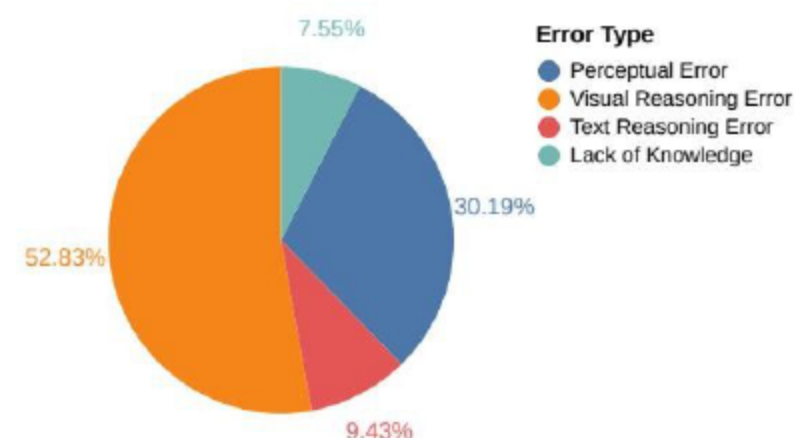
– Broad comparison: humans vs. open and closed MLLMs.

• Scaling test-time compute yields only modest gains.

– Computational costs and generation time analysis.
– Stage 1 has fixed API calls.
– Stage 2 involves iterative review, affecting API calls.
– Generation time longer than PPT output due to VLM-based Page Review.

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Error Analysis, Impact, and Future Work



- Visual reasoning dominates errors (~52.83%).
 - Weak spatial/transformational skills: rotate, flip.
 - Physics failures: magnetism and vectors misread.
- Impact: safety, education, and scientific use.
 - Textual CoT can backfire; train stepwise with grounding.
 - Future: standards and community benchmarks.

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Errors, Impact & Next Steps